Arista Access Switch Node HSR-ACC-27 Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9978

Description

This procedure outlines the steps to troubleshoot and resolve hardware/environmental faults on the Arista Access Switch Node HSR-ACC-27 located at HSR Layout Access. The failure domain is attributed to ASIC memory parity errors, line card crashes, or power supply redundancy loss. In Arista's EOS (Extensible Operating System), the control plane and data plane are decoupled, allowing for more flexible and scalable network designs. However, this decoupling also increases the complexity of fault isolation and resolution. The hardware mechanics involved in this failure domain include the line card's ASIC (Application-Specific Integrated Circuit) memory, which is responsible for processing and storing routing information. When the ASIC memory parity errors occur, it can lead to line card crashes and subsequently affect the overall network availability. Additionally, power supply redundancy loss can also cause the line card to crash, resulting in network downtime. The root cause of this failure domain can be attributed to various factors, including hardware manufacturing defects, environmental factors such as temperature and humidity, or human error during maintenance activities.

Pre-requisite Checks

- Ensure remote hands are dispatched to HSR Layout Access to physically access the node.
- Verify the redundant link status to ensure that the network is not impacted by the failure.
- Check the power supply redundancy to ensure it is functioning properly.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware manufacturing defects	High
Line card crashes due to environmental factors such as temperature and humidity	Medium
Power supply redundancy loss due to human error during maintenance activities	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal linecard detail, Expected Output: Linecard slot information, including ASIC memory parity errors, Description: This command is used to display detailed information about the linecard, including ASIC memory parity errors, which can help identify the root cause of the failure.

Command: show system internal power detail, Expected Output: Power supply redundancy information, Description: This command is used to
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display detailed information about the power supply, including redundancy status, which can help identify the root cause of the failure.

Command: show system internal control-plane detail, Expected Output: Control plane information, including routing table and adjacency information, Description: This command is used to display detailed information about the control plane, including routing table and adjacency information, which can help identify the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the faulty linecard, Command: reboot node 1/1/1, Description: This command is used to reboot the linecard and replace it with a new one.

Step 2: Step: Replace the faulty power supply, Command: reboot node 1/1/1, Description: This command is used to reboot the node and replace the faulty power supply with a new one.

Step 3: Step: Verify the network configuration, Command: show running-config, Description: This command is used to verify that the network configuration is correct and consistent across all devices.

Post-Check Verification

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Command: show system internal linecard detail, Expected Output: Linecard slot information, including ASIC memory parity errors, Description: This command is used to verify that the linecard is functioning properly and there are no ASIC memory parity errors.

Command: show system internal power detail, Expected Output: Power supply redundancy information, Description: This command is used to verify that the power supply redundancy is functioning properly.

Command: show system internal control-plane detail, Expected Output: Control plane information, including routing table and adjacency information, Description: This command is used to verify that the control plane is functioning properly and the routing table and adjacency information are consistent.
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Roll-back Procedures

- Step: Restore the original configuration, Command: load merge , Description: This command is used to restore the original configuration from the backup.

- Step: Reboot the node, Command: reboot node 1/1/1, Description: This command is used to reboot the node to ensure that the changes are rolled back.